

Edge-Deployable TinyML Framework for Breast Cancer Detection with Explainable AI from DCE-MRI

Ali Mohammad Nasir Uddin and Khandoker Momotaz Ferdouse

A Thesis in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

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Fall 2025

Letter of Transmittal

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Subject: Letter of Transmittal.

Dear Sir,

Kindly consider this letter as the formal transmittal of our research paper titled “Edge-Deployable TinyML Framework for Breast Cancer Detection with Explainable AI from DCE-MRI.” We have developed this paper as part of our academic research endeavors.

In this paper, we introduce a novel hybrid framework using a convolutional neural network which is trained and then quantized for edge deployment along with incorporating explainable artificial intelligence for better performance and interpretability. Such a solution provides a viable answer to problems associated with efficiency, deployment, and reliability in medical AI.

We respectfully submit this paper for your kind consideration.

Yours Sincerely,

Ali Mohammad Nasir Uddin

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Student's Declaration

We, Ali Mohammad Nasir Uddin & Khandoker Momotaz Ferdouse, being the students of BCSE (Bachelor of Computer Science and Engineering), belonging to the College of Engineering and Technology (CEAT), International University of Business Agriculture and Technology (IUBAT), hereby submit this report.

The title of this thesis is “Edge-Deployable TinyML Framework for Breast Cancer Detection with Explainable AI from DCE-MRI”. This thesis has been prepared and edited by us.

We would like to certify that we, as the authors of this thesis, are solely responsible for the content and structure of this paper and we declare that to the best of our knowledge, there is no material in this thesis that has been published by another person or accepted for the award of any other degree or diploma of the university or other institute of higher learning.

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Supervisor’s Certification

This is to certify that the thesis entitled “Edge-Deployable TinyML Framework for Breast Cancer Detection with Explainable AI from DCE-MRI” has been carried out by Ali Mohammad Nasir Uddin and Khandoker Momotaz Ferdouse, students of the Bachelor of Computer Science and Engineering (BCSE) program under the College of Engineering and Technology (CEAT) at the International University of Business Agriculture and Technology (IUBAT), under my supervision.

To the best of my knowledge, this thesis is the result of the students’ original research work. The content of this thesis has not been submitted, in whole or in part, for the award of any degree or diploma at this or any other university or institution, except where proper acknowledgment has been made. The work is found to be satisfactory and is submitted for examination with my approval.

I recommend that this thesis be accepted and considered for fulfillment of the requirements of the Bachelor of Computer Science and Engineering degree.

Abdullah Mohammad Sakib

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Abstract

In today's world, breast cancer ranks amongst the top killers amongst females and hence there is a requirement of efficient and accurate diagnostic algorithms which can be deployed in practice. Although Dynamic Contrast Enhanced Magnetic Resonance Imaging (DCE-MRI) provides good diagnosis sensitivity, most current DL-based models are very complex in terms of computation cost which restricts them for usage in edge and low resource healthcare settings. To tackle such issues, in this work, we present our customized and unique design of lightweight CNN-Vision Transformer Hybrid model specifically for detecting breast cancer based on DCE-MRI before contrast administration using TinyML framework. Our algorithm uses CNN architecture for localized texture analysis along with transformers for global context modeling, while ensuring model interpretability through LIME based approach after quantization. Experiments were performed on Duke Breast MRI dataset using balanced slices level splits and post quantization was performed on the model (Dynamic-Range and float16) for enabling edge deployments with size of model being less than 5 MB. Post training quantization results have led us to observe 97.0% and 96.75% testing accuracy (for dynamic-range and float16), respectively, along with high values of Recall, Precision, Specificity, and Area Under Curve (AUC).

Keywords: Breast Cancer, TinyML, CNN-ViT Hybrid, Hybrid CNN-Transformer, Edge AI, MRI

Acknowledgments

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A special note of appreciation goes to our thesis supervisor, Abdullah Mohammad Sakib, for his invaluable advice, encouragement and expert supervision, which greatly contributed to the success of our research. His guidance and unwavering support helped us overcome challenges and refine our work. Lastly, we extend our sincere thanks to all CSE faculty members for their ongoing support, as well as our families and coursemates, whose patience, motivation, and assistance played a crucial role in completing this thesis.

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Chapter 1. Introduction

1.1 Background Study

Breast cancer remains one of the most important issues concerning the health of people throughout the world due to the fact that it contributes significantly to the morbidity and mortality of women worldwide [1]. The accurate and timely diagnosis of tumors plays an important role in ensuring patient survival, as well as the effective management of treatment. Compared with other imaging technologies available nowadays, magnetic resonance imaging enjoys its advantages described above [2].

Dynamic Contrast-Enhanced Magnetic Resonance Imaging (DCE-MRI) proves very useful since it offers information about the functional behavior of the tumors concerning their vascularization and perfusion. However, even the pre-contrast T1-weighted MRI slices offer valuable information in terms of anatomical and structural characteristics that could be used in the diagnosis of the tumors. Pre-contrast MRI imaging technology is strongly recommended when the use of a contrast agent is inappropriate or costly [3]. This explains why there has been much interest in developing technologies for automatically processing pre-contrast MR images.

The improvements achieved through deep learning algorithms have significantly affected the domain of medical image analysis, particularly in diagnosing breast cancer from magnetic resonance imaging images. The convolutional neural network (CNN) model has demonstrated excellent results in discovering local textures and spatial attributes, while the vision transformer (ViT) has been extremely efficient in discovering long-range spatial relationships and global information [4,5]. Various studies have utilized different CNN and ViT models as well as combinations of both [6,7].

Even though the mentioned methods have shown great performance, the bulk of the methods rely on massive neural networks that require lots of memory space, employ complicated floating-point computations, and require the use of GPU devices for performing inferences. Consequently, these methods become very hard to apply for practical application cases in a

clinical setting without enough computational resources [8,9]. This is especially the case when few resources are available.

A number of ideas have been proposed regarding the compression of a model and its quantization. While post-training quantization is capable of reducing the size of the model drastically, the consequences in terms of model interpretability and transparency cannot be overlooked [10,11]. The quantization step renders the use of gradients by techniques such as Grad-CAM and Integrated Gradients useless for explanation purposes for the decision making process [12]. Lack of transparency is a critical obstacle to the adoption of such models, particularly in the health care sector where interpretability is crucial. Lastly, incorporating TinyML in MRI guided breast cancer detection using explainable AI is an area that has yet to be explored fully.

1.2 Research Gap

While great improvements have been proposed to detect breast cancer with deep learning in MRI, yet many challenges remain.

First, the current approaches focus on diagnostic performance without making the system usable. High-accuracy CNN and transformer models require substantial resources which are unlikely to be available in resource-poor clinical settings such as rural medical clinics and mobile screening devices [8,9].

Second, although approaches such as model compression and post-training quantization have been leveraged for smaller models, their effects on interpretability have not been considered adequately. The fact that quantized models lack gradient values makes the traditional approach to interpretability impossible and greatly hinders the trust of clinicians in them [12].

Thirdly, the use of TinyML methods in the case of breast cancer detection using MRIs is relatively unexplored. Most studies do not develop algorithms that are explicitly tailored to work under ultra-low memory and computational capacity conditions; rather, they utilize reduced versions of large pre-trained models [10,11].

Fourth, CNN-ViT hybrid models have proven successful, but there is a need for more research into customized lightweight hybrid models suited to medical image classification from pre-contrast MRI slices. This is because most hybrid models still employ heavy backbone models that make them unsuitable for edge deployment [6,7].

Lastly, the absence of an integrative approach to incorporating quantized deep learning models and model agnostic explainable artificial intelligence approaches exists. It is imperative to ensure the interpretability of quantized models to support clinical decision-making. However, this problem has not been adequately addressed in literature.

1.3 Problem Statement

The core problem addressed in this research is:

What would be a strategy to develop an efficient, precise, and interpretable deep learning architecture for detecting breast cancer using pre-contrast T1-weighted MRI that could be implemented in TinyML?

This broader question involves the following issues:

- **Lightweight Architecture Design:** Building an entirely customized CNN-ViT model architecture that provides high diagnostic accuracy while keeping memory size and computations at the bare minimum.
- **TinyML Compatibility:** Making sure that the model is deployable on devices with limited resources by performing post-training quantization.
- **Quantization Robustness:** Maintaining or enhancing diagnostic capability upon quantization.
- **Explainability Retention:** Allowing explainability of the quantized model despite the lack of gradients.
- **Clinical Trustworthiness:** Ensuring that the model pays attention only to clinically important areas of the breast tissue.

1.4 Research Contribution

This research makes the following key contributions:

1. Custom Lightweight CNN–ViT Architecture:

An entirely customized CNN–Vision Transformer hybrid model tailored for TinyML applications is suggested for the diagnosis of breast cancer from pre-contrast T1-weighted DCE-MRI images. This CNN–Vision Transformer network integrates a lightweight CNN backbone for local feature extraction and custom Vision Transformer modules for global contextual understanding.

2. TinyML Deployment via Quantization:

A new deployment approach, which involves TensorFlow Lite quantization after training, is proposed, featuring dynamic-range and float16 quantization algorithms. The proposed models have a very small size (less than 5 MB) while preserving high diagnostic precision.

3. Explainable AI for Quantized Models:

Explanation via model-agnostic explainable artificial intelligence with LIME technique is incorporated to offer accurate interpretation of quantized models. The solution offers an effective remedy for the shortcomings associated with gradient-based explainability techniques.

1.5 Research Objectives

The primary objectives of this study are:

- The goal is to design an end-to-end custom architecture of CNN–ViT that is highly accurate at diagnosing breast cancer on pre-contrast T1-weighted slices.
- The need is to devise methods for model quantization after training to reduce their size substantially without compromising accuracy.
- The requirement is for interpretability of these quantized models with explainable AI methods.

- The task is to prove that MRI-based breast cancer diagnosis can be done on TinyML devices.

1.6 Thesis Outline

The remainder of this thesis is organized as follows:

- **Chapter 1: Introduction** explains the background, motivation, research gaps, problem statement, research objectives, and research contributions for the study.
- **Chapter 2: Literature Review** explores the existing literature on breast cancer classification using MRI, CNN, ViTs, hybrid models, TinyML, quantization, and explainable AI.
- **Chapter 3: Methodology** provides information on the Duke Breast Cancer MRI dataset, data preprocessing techniques, and data pre-processing pipeline, along with the methodology related to the CNN–ViT hybrid network design, quantization processes, and explainability.
- **Chapter 4: Results** highlights detailed experiments, performance analysis, quantization results, interpretability results, discussions on results, limitations, and clinical/deployment implications.
- **Chapter 5: Conclusion** concludes the study by highlighting the research contributions and suggesting future research directions.

Chapter 2. Literature Review

The study of automated breast cancer detection using MRI techniques has seen tremendous growth in recent times due to the progress made in deep learning techniques and increased access to large amounts of medical imaging data. The initial research work conducted was based on deep CNN models trained using the entire DCE-MRI scans, multiparametric MRI scans, or a combination of T1 and T2 images. These systems showed exceptional results in terms of diagnostic performance but were difficult to implement in practical clinical settings due to excessive computation costs.

A number of surveys and reviews have been conducted to provide an overview of the advancements made in this field. For instance, Abdullah et al. [13] provided a systematic review on the use of deep learning algorithms in breast MRI interpretation, pointing out some of the common architectures, datasets, and metrics employed in research. Though the review proves the efficacy of CNN models, it reveals some of the major challenges, such as scanner differences, poor generalizability, and inadequate focus on lightweight models. Additionally, Adam et al. [14] highlighted some of the latest advances in using deep learning algorithms in detecting breast cancer. They revealed some of the most widely adopted models, including CNN, U-Net, autoencoder, and transformer architectures. However, they also noted some significant drawbacks, such as dataset imbalance, non-standard evaluation protocols, and low emphasis on efficient and lightweight models.

In order to overcome the limitations imposed by the computational limitations, research efforts have been made in the direction of developing efficient CNN models. In a study conducted by Wang et al. [15], it was shown that fine-tuning MobileNets is capable of performing at par with other CNNs for classifying breast lesions in DCE-MRI images while offering significant reductions in both model size and computation costs. However, the downside of employing lightweight CNN models for medical image analysis is that they employ localized receptive fields only and cannot effectively incorporate the context dependency present in anatomically complex images.

While exploring the potential of CNNs, Vision Transformers (ViTs) have attracted interest due to their capability of modeling global relations using self-attention layers. In a comprehensive analysis of ViTs and CNNs on different medical imaging tasks, Takahashi et al. [16] observed that ViTs are highly efficient in modeling long-distance dependency but need significantly more training data and strong regularization to stabilize results. However, such demands create problems for applications in medical imaging due to the scarcity of labeled data. Hybrid models of CNN and Transformer have been developed to address these shortcomings. Jahan et al. [17] designed a hybrid CNN-Transformer approach for breast cancer screening and molecular subtyping, proving that the integration of local features extracted by CNN with global attention of Transformer enhances classification. Likewise, Tagnamas et al. [18] generalized hybrid architectures for multiple ultrasound imaging tasks, performing tumor segmentation and classification in parallel by exploiting complementary feature representations. Despite their superior performance compared to individual CNNs and ViTs, the complex architecture of hybrid models is a computational bottleneck, limiting their application in low-power and edge settings.

The significance of edge computing and TinyML-based healthcare applications has also been recently discussed. Islam et al. [19] employed XAI methods in breast cancer risk assessment models, thereby emphasizing the need for model interpretability for clinical acceptance, especially among under-represented communities. Arthi and Krishnaveni [20] suggested a TinyML-based healthcare assistance framework where model compression and explainability were considered together for enabling energy-efficient predictions. While the research papers cited here may not necessarily discuss MRI-based imaging pipelines explicitly, they contribute to the current understanding that clinical application demands not only accuracy but also model interpretability and efficiency.

are becoming increasingly popular approaches towards shrinking models while maintaining their performance. As shown by Qu et al. [21], int8 PTQ when applied to models trained for 3D medical imaging tasks does not lead to loss in accuracy, suggesting the ability to utilize quantized models even in resource-limited settings. Nevertheless, current work on quantization primarily addresses either CNN models or large 3D models but not the case of quantized hybrid CNN-Transformer models for 2D MRI classification tasks.

As a summary of all the research works mentioned, they give some important knowledge about breast MRI analysis, lightweight CNNs, Transformer models, hybrid models, and TinyML. But there is room for improvements in some areas. First of all, many breast MRI research papers used post-contrast or multiparametric images for MRI breast segmentation and paid little attention to pre-contrast T1-weighted MRI, which could be considered clinically safer due to the same image consistency. Second, most hybrid CNN and Transformer models were not optimized for memory and computational resource requirements.

In order to address the shortcomings mentioned above, this study contributes to the current body of knowledge by designing a completely novel CNN-ViT hybrid model tailored specifically for pre-contrast T1-weighted DCE-MRI. Furthermore, this work aims at enhancing the suitability of the aforementioned framework towards clinical application, specifically as a solution for TinyML-powered deployment of the model. Explainability using LIME is incorporated into the framework, ensuring the trustworthiness of the diagnosis made through the use of the framework.

Ref.	Study	Problem Addressed	Proposed Solution	Key Limitations
[13]	Abdullah et al. (2025)	Lack of consolidated understanding of deep learning methods for breast MRI	Systematic review of CNN- and DL-based breast MRI approaches, summarizing architectures, datasets, and performance metrics	Did not address deployment feasibility, lightweight models, or TinyML constraints; largely focused on accuracy-driven methods
[14]	Adam et al. (2023)	Challenges in breast cancer detection using deep learning across heterogeneous datasets	Comprehensive review of CNNs, U-Nets, autoencoders, and Transformer-based models	Highlighted issues such as dataset imbalance and scanner variability but did not propose solutions for efficiency or deployment
[15]	Wang et al. (2023)	High computational cost of CNN-based breast lesion classifiers	Use of fine-tuned MobileNet variants for DCE-MRI lesion classification	Limited global context modeling; relied solely on CNNs without attention mechanisms
[16]	Takahashi et al. (2024)	Understanding the trade-offs between CNNs and ViTs in medical imaging	Systematic comparison of CNNs and Vision Transformers across imaging modalities	ViTs required large datasets and heavy regularization; not

Ref.	Study	Problem Addressed	Proposed Solution	Key Limitations
				suitable for low-resource settings
[17]	Jahan et al. (2025)	Limited performance of standalone CNNs or Transformers in breast cancer classification	Hybrid CNN–Transformer architecture combining local and global features	High computational complexity; not optimized for edge or TinyML deployment
[18]	Tagnamas et al. (2024)	Need for multi-task learning in tumor analysis	Hybrid CNN–Transformer model for simultaneous tumor segmentation and classification	Focused on ultrasound imaging; model size and power consumption not addressed
[19]	Islam et al. (2024)	Lack of interpretability in breast cancer risk prediction models	Integration of explainable AI (XAI) techniques for regional risk prediction	Did not directly address MRI-based imaging or model compression
[20]	Arthi and Krishnaveni (2024)	Energy inefficiency of healthcare ML systems	TinyML-based framework combining compression and explainability	Imaging data not considered; applicability to MRI not validated
[21]	Qu et al. (2025)	Large memory footprint of medical imaging models	Post-training quantization (int8) applied to 3D medical models	Focused on CNN-based 3D models; hybrid CNN–Transformer quantization not explored

Table 2.1: Comparative Analysis of Related Works in Breast Cancer Imaging and TinyML

Chapter 3. Research Methodology

In this section, the framework based on TinyML has been presented for detecting breast cancer from pre-contrast DCE-MRI images. The framework has five core components of methodology: dataset preprocessing, model design, training approach, TinyML conversion by using post-training quantization, and integration of explainability.

3.1 Overview

For an input pre-contrast T1-weighted MRI slice $x \in \mathbb{R}^{H \times W \times 3}$, wherein the original grayscale image is replicated three times across the channels to allow for compatibility with CNN and Transformer based models, the proposed pipeline generates a binary prediction $\hat{y} \in [0,1]$, indicating the presence/absence of a tumor [22].

The pipeline first consists of a standardized preprocessing and augmentation step, wherein the input image is converted as $x' = A(x)$. This ensures robustness and generalization while maintaining anatomical plausibility. This augmented image is then fed to a feature extraction backbone network, formulated as $Z = f_{\theta}(x')$, where f_{θ} refers to one of the networks tested, which include MobileNetV2, MobileNetV3Small, a Vision Transformer (ViT), a MobileNet-ViT fusion model, or the proposed CNN-ViT model.

The resultant feature map Z is further fed to a classification head, which performs global average pooling on the feature maps and passes it through a sigmoid activated linear layer to generate the output prediction,

$$\hat{y} = \sigma(w^T \text{GAP}(Z) + b),$$

where w and b denote the learnable classifier parameters and $\sigma(\cdot)$ is the sigmoid function. In order to make sure that the deployment process takes place under severe resource limitations, quantization is performed on the trained model in two different ways, namely dynamic range quantization and float16 precision, thus creating TinyML-ready versions of the models. Finally, model-agnostic explainability is implemented through Local Interpretable Model-

agnostic Explanations (LIME). This technique offers visual explanations of local image regions that have played an important role in deciding the output of the model [23].

3.2 Dataset and Preprocessing:

The current study employs the publicly available Duke Breast Cancer MRI Dataset, sourced from the **Cancer Imaging Archive (TCIA)** [24]. This is an extensive multi-center database of DCE-MRI exams performed on patients with varied medical histories. While the dataset offers complete temporal information in terms of pre- and post-contrast scans, the current study only considers pre-contrast T1-weighted MRI slices. The rationale behind this decision can be explained by several factors, including the retention of essential anatomical information necessary for detecting malignancies, the absence of reliance on contrast agent absorption, and uniform intensity distribution, which makes it more appropriate for lightweight TinyML inference. Additionally, abstaining from contrast agents, especially gadolinium-containing compounds, poses fewer risks to patient health, especially those suffering from kidney diseases, lowers acquisition costs, and ensures compliance with practical implementation scenarios.

Data used for each instance within the dataset includes axial T1-weighted MRI slices without contrast, lesion detection using bounding boxes, and metadata about lesion properties and patient demographics. These elements ensure proper image slice selection, accurate lesion location, and correct classification of instances. In all, 300 patient data sets were considered for this experiment resulting in a total of 4,000 training instances consisting of an equal number of tumor-positive and negative cases, and a similar number of validation and test images each totaling 400 images.

In terms of slice extraction and labeling, all slices of pre-contrast MRI scans were extracted for each subject. By using the bounding box annotations provided, tumor-containing slices were determined to enable building the binary classification dataset. In particular, positive slices are those containing lesion bounding boxes; however, each bounding box was carefully inspected to make sure that there is actually tumor tissue. For negative slices, they are selected from slices that are at least 8 positions away from the nearest tumor-containing slice to avoid boundary regions where lesion remnant might appear in. All of the negative slices were examined to check that there are no structural abnormalities or suspicious findings on the

images. This method of slice-level labeling enables building a binary classification dataset with well-separated classes and low contamination between 5 classes.

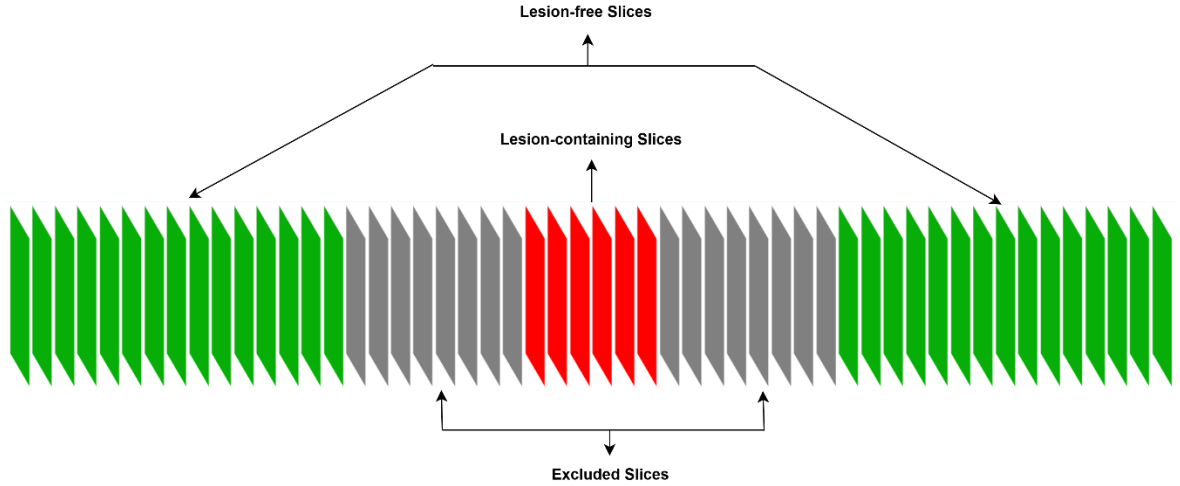


Figure 3.1: The slice-level class labeling approach employed for creating the dataset is explained below. The positive class samples include tumor-infected slices that have been detected based on bounding boxes supplied and confirmed to contain tumors. The negative class samples have been chosen from slices at least eight places apart from any tumor-infected slice.

3.2.1 Dataset Balancing:

Breast MRI images containing tumors will occur less frequently in datasets than images without tumors, resulting in a notable imbalance of the classes in the dataset, which may skew learning towards the majority class. To address this problem, the dataset was balanced by including the same number of images with and without tumors in the dataset. Class balancing ensures that learning equally weighs both classes, leading to better generalization performance, convergence, and avoids biased decision boundaries, which is highly relevant for medical diagnosis applications, where missing tumors may have dire consequences.

3.2.2 Image Preprocessing Pipeline:

A consistent and standardized preprocessing protocol was used to process all of the MRI slices to be able to use the data across convolutional neural networks, vision transformers, the fusion architecture, and the novel CNN-ViT model. The preprocessing protocol aimed at retaining clinically meaningful information while ensuring that the inputs were ready for training and TinyML. The preprocessing pipeline is comprised of the following stages:

1. Resizing and Channel Replication: The MRI slices were resized to a uniform size of 512×512 pixels with three channels. While the MRI images were initially in a single-channel format, all slices were duplicated in three channels to adhere to the data specifications expected by both CNNs and Transformers. The decision to employ a high-resolution spatial dimension of 6 was deliberate since it enabled the model to preserve intricate anatomical and textural characteristics, including lesion borders and heterogeneity, critical in classifying tumors within breast MRI [25].

2. Normalization: The normalization process was carried out by means of two different techniques which had been specially designed for use in the specific architecture. This concerned the cases of MobileNetV2 and CNN-ViT, where the pixel values were directly normalized to lie in the interval $[0,1]$. On the contrary, MobileNetV3, ViT, and the combination of MobileNet and ViT models used the built-in TensorFlow preprocessors, which were consistent with the original architecture of the network.

3. Data Augmentation: In order to improve robustness and avoid overfitting, controlled data augmentation was only performed on the training data [26]. The augmentations involved flipping images horizontally and vertically, rotating images by a small angle between $\pm 18^\circ$, zooming images slightly between $0.9\times$ and $1.1\times$, and altering image contrast randomly by $\pm 10\%$. Care was taken to ensure the augmentation remained anatomically plausible and tumor shape consistent while also creating natural variations caused by differences in scanner use, patient position and acquisition condition.

3.2.3 Final Dataset Structure:

After preprocessing, balancing, and annotation steps, the resulting dataset had two well-separated classes: the first class comprised the images with tumors present, while the second class had no tumors at all. The dataset was split into training, validation, and testing sets with ratios of 80%, 10%, and 10%, respectively. Significantly, the balance between classes was maintained throughout the splits in order to guarantee stable training dynamics, unbiased tuning of hyperparameters, and correct evaluation of the network performance on new data. This structured dataset organization provides a solid foundation for a fair comparison of the baseline models, fusion variants, and proposed CNN-ViT hybrid architecture.

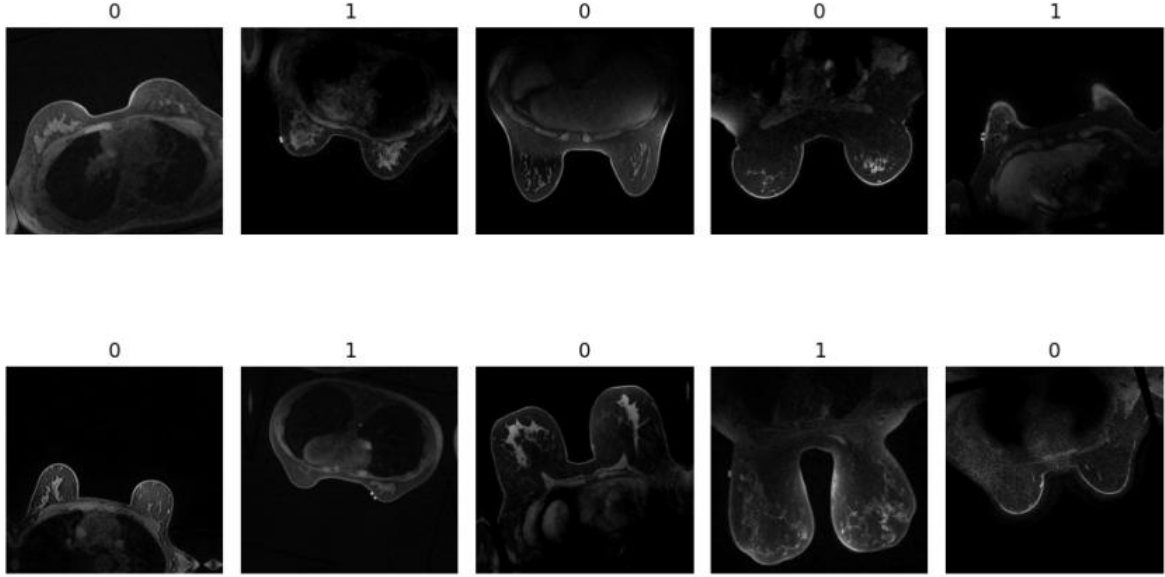


Figure 3.2: Examples drawn from the breast cancer MRI database applied in this research work: T1-weighted pre-contrast MRI slices of positive (1-tumor present) and negative (0-tumor absent) instances. These instances showcase the presence of tumor within the database and thus emphasize the necessity of adopting a lightweight, explainable, and deployable framework for accurate breast cancer identification.

No patient-level balancing or preprocessing that was temporal and contrast-based was considered, as the goal was to maintain the characteristics of pre-contrast MRI and to ensure a realistic distribution in clinical imaging. These preprocessed data were used for training the CNN, ViT, and hybrid models, as well as Post-Training Quantization (PTQ) for optimizing the models to work with TinyML.

3.3 Model Architecture:

The following section explains the architecture of the lightweight CNN-ViT Hybrid Model that is employed in this research to create a breast cancer detection model that can be deployed at the edge-devices. The architecture of the proposed model was developed to ensure maximum accuracy while keeping the computational load, parameters, and memory low to satisfy **TinyML** constraints.

3.3.1 Rationale for a CNN-ViT Hybrid Architecture:

In breast MRI images, both high-level global context patterns (such as bilateral symmetry and structural patterns) and low-level detailed local patterns (such as lesions' borders and texture heterogeneity) can be found. Without global receptive fields, Convolutional Neural Networks (CNNs) can extract local spatial patterns effectively with hierarchical filters [27]. However, Vision Transformers (ViTs) are good at handling long-range dependency modeling via self-attention mechanisms. Nevertheless, ViTs often need large amounts of data for training and have higher computational costs [28].

In this study, we propose an architecture-specific lightweight hybrid CNN-ViT tailored for classification tasks in breast DCE-MRI T1-weighted pre-contrast imaging to exploit the benefits offered by both approaches. In this regard, whereas the encoder captures the long-range dependencies across the entire image, the CNN backbone captures local information useful for diagnosis. Most importantly, our proposed hybrid framework achieves remarkable gains in discriminative capacity without compromising model compactness and TinyML compliance. The novel framework is entirely self-learned without any pretrained backbones, making it suitable for MRI intensity distributions.

3.3.2 Dataset and Preprocessing Pipeline:

In the model architecture, there is a tight integration between the data handling procedure and the training process itself to achieve optimum efficiency and uniformity during training. All input samples have been directly taken from the preprocessed dataset discussed previously in the paper, employing the native functionality provided by TensorFlow for data loading. The data handling process assigns class labels according to the folder structure used, and batch and shuffle are done on-the-fly, along with prefetching.

In terms of architecture, the input dimension remained constant in terms of spatial size at 512x512 pixels with three channels so that the fine-grained texture and structural information necessary for precise breast MRI tumor classification could be preserved, while keeping architectural homogeneity throughout the hybrid network. Such a consistent input scheme enables the proposed hybrid architecture to effortlessly integrate convolutional feature extraction with attention through transformer modeling, without additional adaptation layers.

The lightweight transformations needed for training, such as normalization and augmentation, were included directly in the input pipeline to conserve memory space and eliminate the need for any repeated processing of the dataset before training. These transformations were done to ensure stable optimization and generalization during the evaluation process. Lastly, the input pipeline incorporated caching and prefetching techniques to reduce any I/O bottleneck issues and optimize GPU usage during training.

3.3.3 Proposed CNN-ViT Hybrid Architecture:

The suggested model has a hybrid architecture where the convolutions and the inductive bias associated with them can be used to combine with global context modeling using transformers. The architecture has a total of five main components, namely, the convolutional feature extractor, the patch embedding layer, the tokenization process involving the learning of the classification token and the position embeddings, the transformer encoder, and finally the classification head.

1. Convolutional Feature Extractor:

The custom lightweight convolutional backbone is used to extract low-level and mid-level spatial features. The architecture is designed using four layers with residual skip connections, where each layer contains Conv-Batch Norm-RELU layers along with max-pooling operations for spatial down sampling. The number of filters is gradually increased from 32, 64, 128, to 256 in each subsequent layer. The hierarchical architecture enables the network to gradually increase the receptive field while retaining the spatial features at finer scales. The backbone output is generated as a compact but semantically rich feature map with dimensions 32 x 32 x 256, serving as an input to the transformer model.

2. Patch Extraction and Linear Embedding:

In order to establish a link between the convolved image representation and the transformer, the output feature map undergoes an extraction process, and the resultant features are divided into a set of disjoint patches through the Patch Extractor layer, which consists of extracting patches using TensorFlow's extract patches function. If the feature map size is $H \times W$, and the patch size is $P \times P$, so:

$$N = \frac{H}{P} \times \frac{W}{P} \dots\dots\dots(1)$$

Each patch is flattened and projected into a fixed-dimensional embedding space of size 128 using a learnable linear projection:

$$Z_0 = W_{\text{proj}} X_{\text{patch}} \dots\dots\dots(2)$$

This projection aligns the patch representations with the transformer's token-based input format while maintaining computational efficiency.

3. CLS Token and Positional Embedding:

To allow learnable classification of the token [CLS], this token is added to the front of the sequence of patches embedded in order to aid global feature learning. The idea here is for this token to capture information about all other patch tokens using self-attention mechanism. Spatial information can be retained by adding position embeddings of the token sequence:

$$Z = [Z_{\text{CLS}}, Z_0] + E_{\text{pos}} \dots\dots\dots(3)$$

where E_{pos} denotes the learnable positional embedding matrix. The positional embeddings are dynamically indexed to match the effective sequence length during training and inference.

4. Transformer Encoder:

The output sequence is then passed through a stack of three Transformer encoder modules where pre-layer normalization architecture is used to ensure stability during optimization. In each encoder block, self-attention with multiple attention heads, four in this case, is used, allowing the model to learn different kinds of relationships in separate subspaces within the embedding space. Following the attention mechanism is a position-wise feed forward network made up of two fully-connected layers with an intermediary dimension of 256 and GELU non-linearity. Dropout regularization and residual connections are used in all encoder modules to reduce overfitting and encourage generalization.

The output of a single Transformer encoder block can be expressed as:

$$Y = \text{FFN}(\text{MHA}(\text{LN}(X))) + X \dots\dots\dots(4)$$

where X is the input token sequence, $\text{LN}(\cdot)$ denotes layer normalization, $\text{MHA}(\cdot)$ is multi-head self-attention, $\text{FFN}(\cdot)$ is the position-wise feed-forward network, and the residual connection is added back to preserve gradient flow.

5. Classification Head:

After transformer encoding, the output associated with the [CLS] tag alone is kept as the compact global feature vector. This feature vector is then subjected to classification head, which comprises of the following layers: Layer Normalization, two fully connected layers having 128 and 64 neurons respectively, and dropout. Finally, a dense layer followed by sigmoid activation gives the output of binary classification. L2 regularization is also used in the fully connected layers to improve generalization. The final probability output for binary classification is given by:

$$y^{\wedge} = \sigma(W_2 \text{ReLU}(W_1 \text{LN}(Z_{\text{CLS}}))) \dots \dots \dots (5)$$

where Z_{CLS} is the CLS token representation after transformer encoding, $\text{LN}(\cdot)$ denotes layer 10 normalization, W_1 and W_2 are the learnable weights of the fully connected layers, and $\sigma(\cdot)$ is the sigmoid activation function producing the probability of the positive class. The full hybrid model contains approximately 2.3M parameters, making it fully compatible with TinyML deployment after quantization.

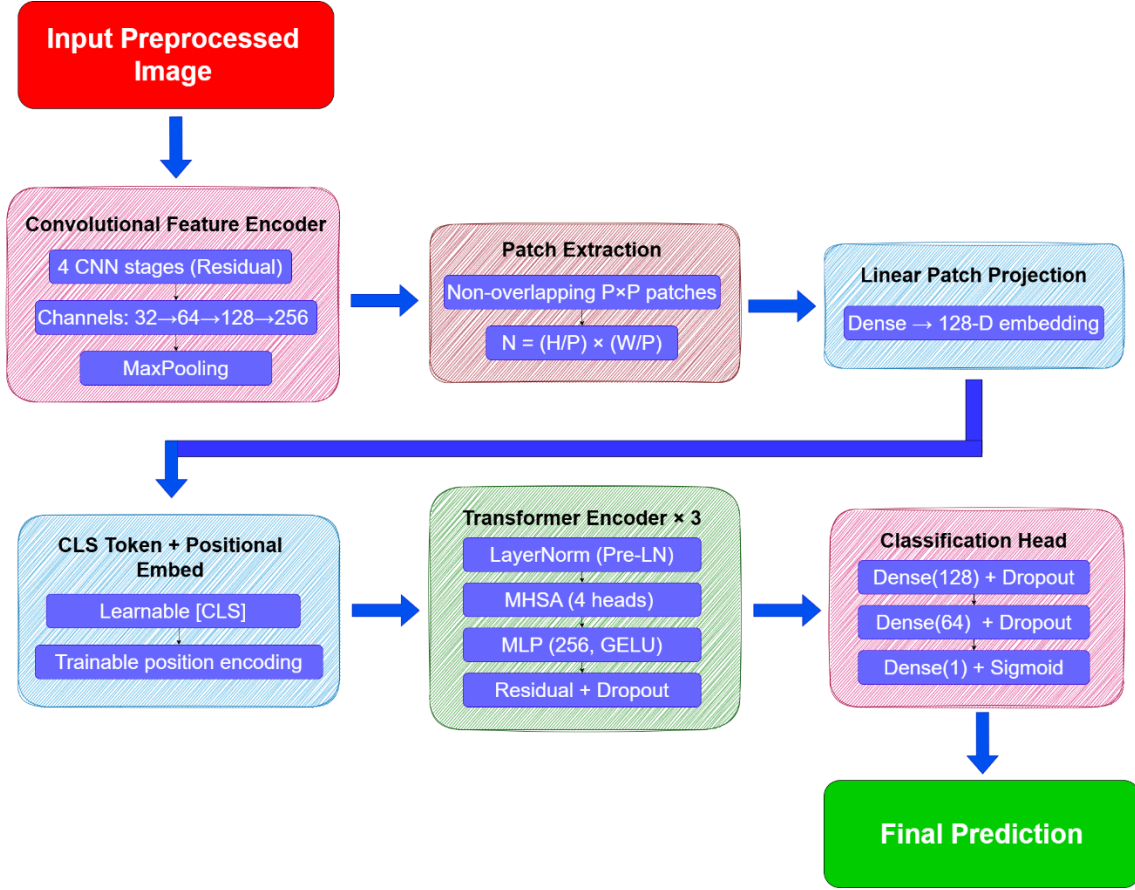


Figure 3.3: Overview of CNN-ViT architecture. The proposed approach begins by performing spatially localized feature extraction via a lightweight customized CNN architecture, consisting of convolution-batch normalization layers. The output is subsequently downsampled to fixed-size patches used in the Vision Transformer network that executes global attention to learn long-range correlations in the dataset. The two networks' outputs are merged in a joint feature space followed by a classifier.

Training Strategy:

A warmup-cosine learning rate schedule was adopted to stabilize early training and promote smooth convergence. Let s denote the current training step.

- **Warmup Phase:** During warmup:

$$\eta(s) = \eta_0 \cdot (s / S_{\text{warm}}) \dots \dots \dots (4)$$

where $\eta_0 = 3 \times 10^{-4}$ is the base learning rate, and S_{warm} is the number of warmup steps (minimum one full epoch).

- **Cosine Decay Phase:** After warmup, the learning rate follows cosine decay:

$$\eta(s) = \eta_0 [\alpha + (1 - \alpha) \frac{1}{2} (1 + \cos(\pi \cdot (s - S_{\text{warm}}) / (S_{\text{total}} - S_{\text{warm}})))] (5)$$

where $\alpha = 10^{-4}$ defines the minimum learning rate fraction and S_{total} is the total number of training steps.

- **Optimization and Regularization:** The model was optimized using the Adam optimizer with binary cross-entropy loss and label smoothing (with $\epsilon = 0.03$). Both CNN and Transformer parts were subjected to early stopping (patience = 10) and model checkpointing in order to keep the best performing model.

Algorithm: CNN–ViT Hybrid Forward and Training Pipeline with Preprocessing Require:

Input MRI slice $X \in \mathbb{R}^{H \times W \times 3}$, label y

Ensure: Trained CNN–ViT hybrid model with quantized variants and explainability

1. Data Preprocessing and Augmentation:

(a) Resize: $X_{\text{resized}} \leftarrow \text{Resize}(X, 512 \times 512)$

(b) Normalize intensities:

$$X_{\text{norm}} \leftarrow \text{Normalize}(X_{\text{resized}})$$

(c) Apply random augmentations:

$$X_{\text{aug}} \leftarrow A(X_{\text{norm}}) \text{ (flip, rotation, zoom, contrast)}$$

2. Convolutional Stem:

(a) $F_0 \leftarrow \text{ReLU}(\text{Conv}_{3 \times 3}(X_{\text{aug}}, 32))$

(b) $F_1 \leftarrow \text{BN}(\text{Conv}_{3 \times 3}(F_0, 32))$

(c) $F_2 \leftarrow \text{MaxPool}_{2 \times 2}(F_1)$

(d) Apply residual blocks with increasing channels: 32, 64, 128, 256

(e) $F_{\text{CNN}} \leftarrow \text{MaxPool}_{2 \times 2}(\text{last residual output})$

3. Patch Embedding:

(a) $E \leftarrow \text{PatchExtractor}(F_{\text{CNN}})$

(b) $Z_0 \leftarrow \text{AddClsToken}(E) + \text{PosEmbedding}$

4. Transformer Encoder Blocks (4 layers):

(a) For each layer $l = 0, 1, 2, 3$:

i. $H_l \leftarrow \text{LN}(Z_l)$

ii. $A_l \leftarrow \text{MHSA}(H_l, H_l)$

iii. $U_l \leftarrow Z_l + A_l$

iv. $H_2 \leftarrow \text{LN}(U_l)$

v. $M_l \leftarrow \text{MLP}(H_2)$

vi. $Z_{l+1} \leftarrow U_l + M_l$

5. Global Representation:

(a) $Z_{\text{final}} \leftarrow \text{LN}(Z_4)$

(b) $h \leftarrow \text{FlattenOrGAP}(Z_{\text{final}})$

6. Classification Head:

$$\hat{y} \leftarrow \sigma(\text{Dense}(h, 1))$$

7. Training:

$$L = -y \log \hat{y} + (1-y) \log (1 - \hat{y})$$

Update parameters using chosen optimizer.

8. PTQ:

(a) Apply Dynamic-Range PTQ

(b) Apply Float16 PTQ

9. Explainability using LIME:

(a) Fit local surrogate model: $g = \arg \min_{g \in G} L(f, g, \pi_x) + \Omega(g)$

(b) Highlight regions most influential to model decision

10. Return: Trained and quantized CNN–ViT hybrid model with LIME explanations.

3.4 Experimental Setup:

All experiments were conducted on a Kaggle GPU environment using identical training, validation, and test partitions to ensure fair comparison. Each model was trained under the following conditions:

- Batch size: 16
- Image size: $512 \times 512 \times 3$
- Epochs: up to 100 with early stopping.
- Preprocessing:

identical pipeline including resizing, augmentation, and normalization

Baseline comparisons included:

- Pretrained CNN
- Custom ViT
- MobileNet-ViT fusion model
- CNN-ViT hybrid (proposed)

All models were evaluated in both FP32 and quantized forms

3.5 TinyML Architecture:

The TinyML approach involves deploying ML models on extremely low-power edge devices with low memory, computation ability, and energy consumption constraints [29]. In order to cope with such constraints, we developed a lightweight ML pipeline for pre-contrast breast MRI classification. The pipeline incorporates different efficient models, such as MobileNetV2, MobileNetV3Small, our own Vision Transformer (ViT), CNN+ViT fusion model, and the proposed CNN+ViT hybrid model. We have selected these models because of their small size and good performance-to-cost ratio [30-32].

To reduce the memory footprint even more, we considered two types of post-training quantization (PTQ): dynamic range quantization and float16 quantization. Dynamic range quantization involves conversion to 8-bit integer weights but leaves activations in floating point, while float16 quantization decreases weight precision to FP16 [33]. If W_{32} represents full precision weights and $Q(\cdot)$ represents quantization, then dynamic range PTQ can be represented mathematically as:

$$W_8 = Q_{\text{int8}}(W_{32}), y^{\wedge} = f(x; W_8) \dots \dots \dots (6)$$

and float16 PTQ as:

$$W_{16} = Q_{\text{fp16}}(W_{32}), y^{\wedge} = f(x; W_{16}) \dots \dots \dots (7)$$

Model quantization brought down the size of models to around 1.06 to 4.91 MB, which made them suitable for TinyML. All the sizes of models fall within the range of less than 5 MB size, making it feasible for TinyML applications [33]. The overall framework demonstrates how aggressive quantization enables accurate MRI-based classification under extreme resource constraints.

TensorFlow Lite Post-Training Quantization (PTQ) was utilized to reduce the size of the fully trained network to enable efficient implementation of the proposed CNN-ViT hybrid architecture in constrained TinyML applications [34,35]. The original FP32 network model had a footprint of approximately 26.78 MB, which made it unsuitable for edge devices. The two types of PTQ included dynamic-range quantization and float16 quantization.

- **Dynamic-Range Quantization:**

In this configuration, activations remain in the floating-point data type during inference of the network mat; however, all weights are quantized from 32-bit floats to 8-bit integers [33]. Suppose that W_{fp32} is a tensor of weights. This quantization is carried out using:

$$W_{int8} = Q_{int8}(W_{fp32}) = \text{round}(W_{fp32} / s)$$

where s is a learned scale factor. This strategy greatly conserves memory usage while not needing any calibration information and training. The file size was shrunk from 26.78 MB to 2.38 MB after range dynamic quantization, an almost 11.2× reduction.

- **Float16 Quantization:**

To further explore precision–efficiency trade-offs, we also applied float16 quantization, in which all FP32 weights are cast to 16-bit floating point:

$$W_{fp16} = Q_{fp16}(W_{fp32}).$$

This approach allows maintaining a higher dynamic range than integer quantization [33], which could help those transformer blocks or attention layers that have requirements concerning numerical precision. With regard to float16 quantization, the size was 4.48 MB, yielding a 5.9× reduction relative to the FP32 baseline.

- **TinyML Suitability:**

Both the quantization methods achieved memory usage far below what is generally considered as the limitations of TinyML applications (less than 5 MB). Despite the extreme compression, the models remained stable in performance during inference, thus confirming the feasibility of using the proposed hybrid approach in a low-energy consumption environment at the edge of the network. The findings indicate that the model could be effectively used for diagnostic purposes with TinyML devices while retaining its performance level.

3.6 Explainable AI (XAI) Method:

Deep learning models used in healthcare should be interpretable in order to inform clinical decision making. Nevertheless, most current interpretable solutions are based on the gradients of the model, but such gradients become inaccessible after extensive post-training quantization. Grad-CAM and SHAP approaches both use model gradients:

$$\nabla_x f(x; W) \dots \dots \dots (8)$$

but in quantized TensorFlow Lite models, the gradient path is removed, making these techniques unusable. Therefore, a gradient-free XAI method was required [36].

In order to overcome this limitation, Local Interpretable Model-Agnostic Explanations (LIME) are used by applying perturbation on the input image as well as fitting a local surrogate model to the data. For an input x , LIME generates perturbations x' and optimizes a local model g such that:

$$g = \arg \min L(f, g, \pi_x) + \Omega(g) \dots \dots \dots (9) \quad [g \in G]$$

where π_x measures proximity to x , L enforces local fidelity, and Ω regularizes model complexity.

LIME is independent of the model used and does not depend on gradients; hence, it can be used in quantized MR classification. Nevertheless, it presents one major drawback: LIME sometimes draws attention to areas with bright fatty tissue or similar high intensity tissues neighboring the tumors instead of focusing on the tumors. This is attributed to the fact that LIME uses superpixels [37].

Despite this limitation, LIME provides a practical interpretability tool for TinyML settings, enabling clinical users to inspect model behavior even when gradients are unavailable.

3.7 Research Workflow Summery:

The research is based on the linear pipeline in which it starts with the collection of the pre-contrast MRI images. The pipeline involves a preprocessing step, which is aimed at ensuring uniformity of the collected data. The preprocessing includes 512×512 resizing, normalization, and data augmentation techniques such as rotation and zooming of the dataset.

The key element of the research work is CNN-ViT Hybrid Model, where the convolutional encoder is first used for extracting features, followed by the Transformer-based sequence

processing unit which learns global context information. After training, the model is then subjected to Post-Training Quantization (PTQ) technique in Float16 and dynamic range quantization methods to achieve efficient performance. The entire process culminates with two-level validation:

- **Quantitative:** Performance is validated via metrics like F1-score and Accuracy.
- **Qualitative:** IME provides clarity about the clinical aspects of the algorithm using heat maps to support the classification process.

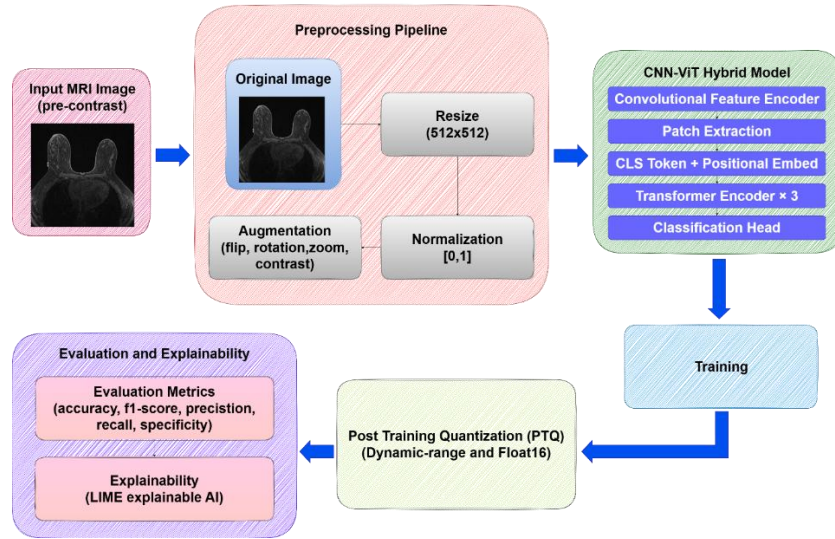


Figure 3.4: Architecture of the system used for breast MRI classification using a deep learning algorithm that is hybrid in nature and trained by post-training quantization.

Chapter 4. Results

4.1 Evaluation Metrics and Confusion Metrics:

In order to conduct an effective assessment of the proposed CNN-ViT model hybrid, a variety of measures was used such as evaluation measures and visualization and threshold-based measures. This is due to the clinical significance of breast cancer classification; therefore, basing one's decision on accuracy alone would be misleading conclusions [38].

4.1.1 Evaluation Metrics

To gain insight into the model behavior, we evaluated standard classification metrics:

Overall Accuracy: This indicates the proportion of correctly classified samples relative to all test instances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \dots\dots\dots (10)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Precision, Recall, F1-Score, and Specificity:

These metrics were computed for evaluating the class-wise diagnostic performance. Precision is a measure of the trustworthiness of the positive prediction, while recall (also known as sensitivity) indicates how well the model can classify tumor positive cases. Specificity measures how accurately the non-tumor cases have been classified, and F1-score gives a harmonic trade-off between both the precision and recall. The metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \dots\dots\dots (11)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \dots\dots\dots (12)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \dots\dots\dots (13)$$

$$\text{F1-Score} = 2 \cdot \left(\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \right) \dots\dots\dots (14)$$

4.1.2 Visualization and Threshold-Based Metrics:

In order to interpret the results of the models further, the following visualization methods were used:

- **Confusion Matrix:** The confusion matrix was analyzed to discover any particular trends in the process of misclassification (for instance, borderline instances being classified either as positive or negative).
- **Receiver Operating Characteristic (ROC) and Area Under the Curve (AUC):** The ROC curve was generated using the one-vs-all approach for all classes, and their respective AUCs were calculated to measure the discrimination ability of the model among the classes. The ROC plots give further information about sensitivity versus specificity, which is important for medical purposes.

4.2 Training and Validation Performance

Training of the CNN-ViT model continued for another 100 cycles using early stopping and retraining. In this regard, both the training accuracy and validation accuracy were seen to show a smooth convergence, whereas the training loss and validation loss were noted to reduce significantly with no sign of overfitting during training. This indicates that the CNN-ViT model was successful in extracting highly robust and discriminative features from the pre-contrast MRIs for breast cancer screening.

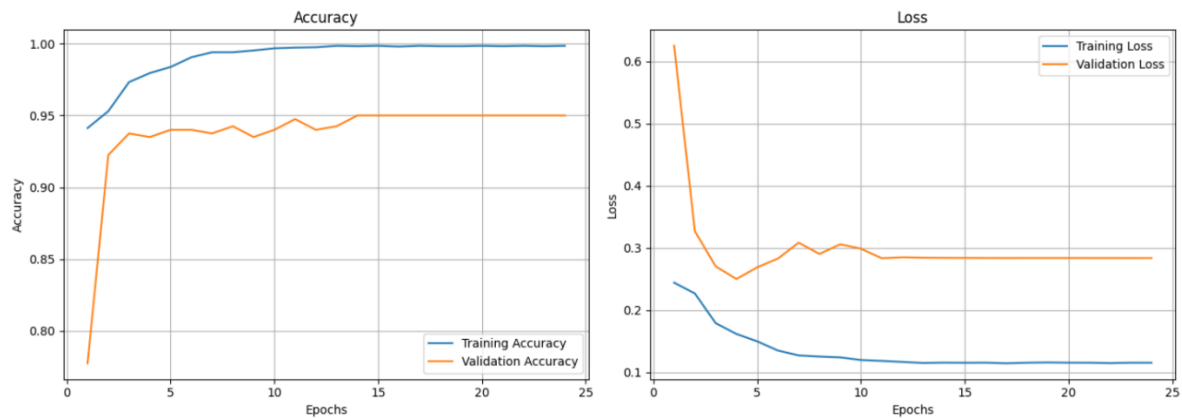


Figure 4.1: Training and validation accuracy for training and validation of the CNN-ViT hybrid model during the first 100 epochs. The graph clearly shows that there is smooth convergence and decreasing loss values, suggesting that learning of the discriminative features from the pre-contrast MR images has been effectively done. However, after 100 epochs, re-training was carried out.

4.3 Quantitative Results:

The effectiveness of the designed CNN-ViT model was evaluated through a series of extensive quantitative analyses using the test dataset. Three different variations of the model were used, including full-precision (FP32), dynamic-range quantization (DQR), and Float16 quantization. A range of evaluation criteria were applied to measure various attributes of classification accuracy, including overall accuracy, F1-score, precision, recall, specificity, and AUC..

4.3.1 Overall Performance Metrics:

The performance of the CNN-ViT hybrid architecture for these three cases is shown in Table 1, which includes FP32, Dynamic-Range Quantization (DQR), and Float16 models. It is clear from the table that DQR gave the best accuracy with 97%, with balanced performance in terms of the remaining parameters, thus proving that there was no need for model degradation when applying quantization techniques. The model in FP32 had an accuracy rate of 96.5%, with improved recall value, and Float16 also provided similar results with 96.75% accuracy rate.

Model	Accuracy	Precision	Recall	F1-Score	Specificity	AUC
FP32	96.5%	95%	98%	96%	95%	98%
DQR	97%	97%	97%	97%	97%	98%
Float16	96.75%	95%	98%	96.79%	95%	98%

Table 4.1: Overall performance metrics of the CNN-ViT hybrid model in FP32, Dynamic-Range Quantization (DQR), and Float16 configurations.

4.3.2 Confusion Matrices

Confusion matrices were also generated to gain an even deeper understanding of the classifying behavior of each of the three models (FP32, DQR, Float16). The information contained in these matrices helps determine the kind of errors the model makes as well as whether the classification mistakes happen between clinically similar classes. It was proven through

confusion matrices that the majority of classification mistakes were happening between adjacent classes.

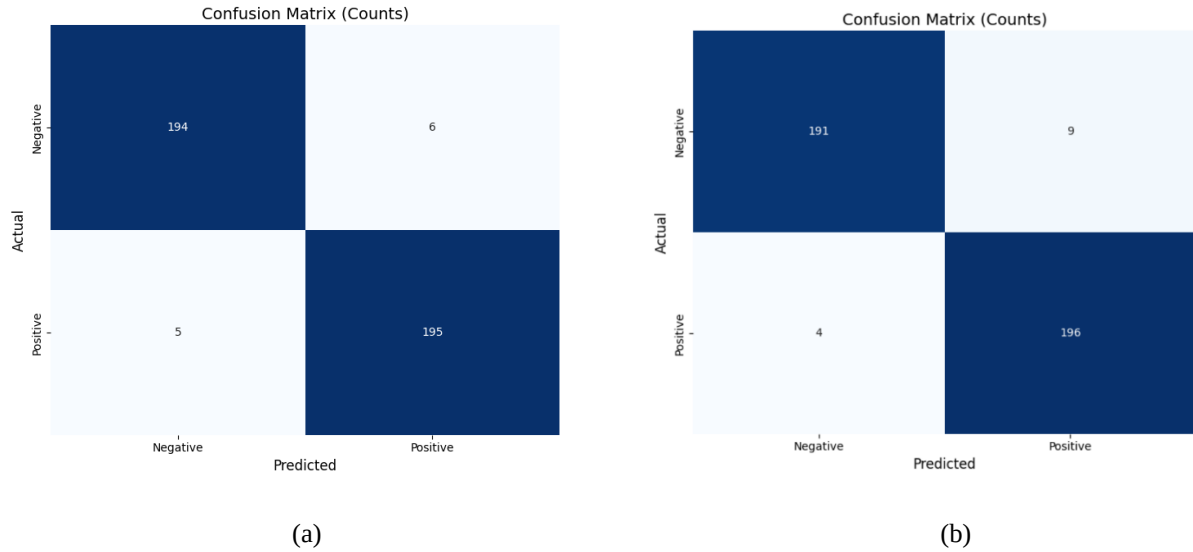


Figure 4.2: Confusion matrices for the hybrid CNN-ViT model after post-training quantization. Figure (a) refers to the Dynamic Range Quantization (DRQ) model, whereas Figure (b) pertains to the Float16 quantized model, showing reliable tumor and non-tumor classification in both cases.

4.3.3 Overall ROC Curve:

Receiver Operating Characteristic (ROC) analysis technique was applied to evaluate more accurately the discriminative capability of the CNN-ViT hybrid quantization model. The ROC curve graphically presents the relationship between the TPR and FPR at different operating points, which provides a threshold-independent evaluation of the model's performance. This is especially critical for medical image processing applications since the operating point will be determined based on the specific requirements of each case [39].

Area Under ROC curve (AUC) acts as an assessment metric of separability between two classes [40]. The greater the value of AUC, the better the discriminatory power of the classifier. This is regardless of the difference between class frequencies or the choice of decision boundary. In this work, ROC curves were constructed for DQR and Float16 quantized models to determine if quantization led to any loss in discriminatory power. The ROC curves reveal that the quantized versions retain the same ROC properties as the original FP32 network.

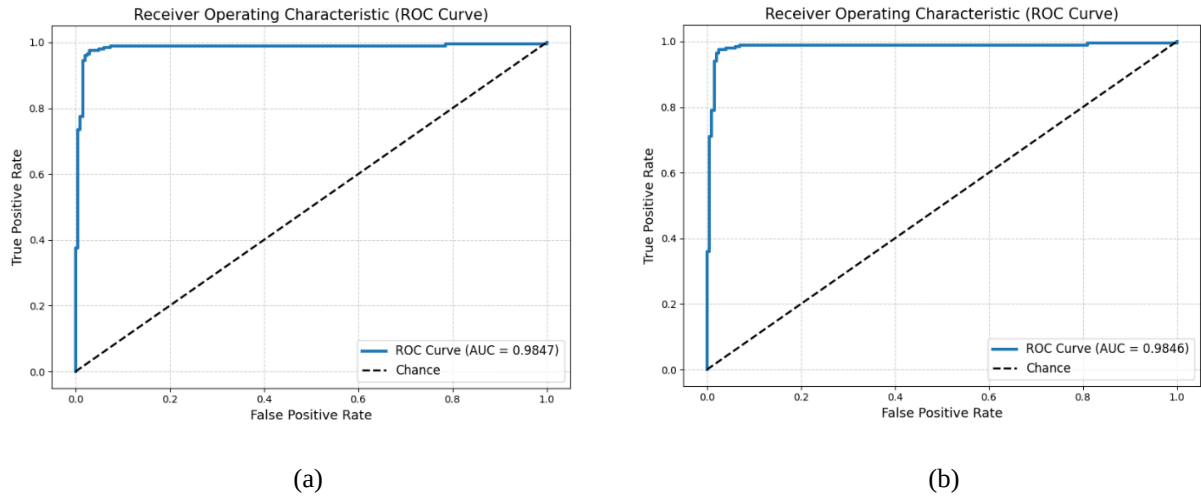


Figure 4.3: The ROC curves of the newly designed hybrid CNN-ViT network after the post-training quantization process. Figure (a) is the ROC curve for the DRQ model, whereas figure (b) represents the ROC curve of the Float16 quantized version.

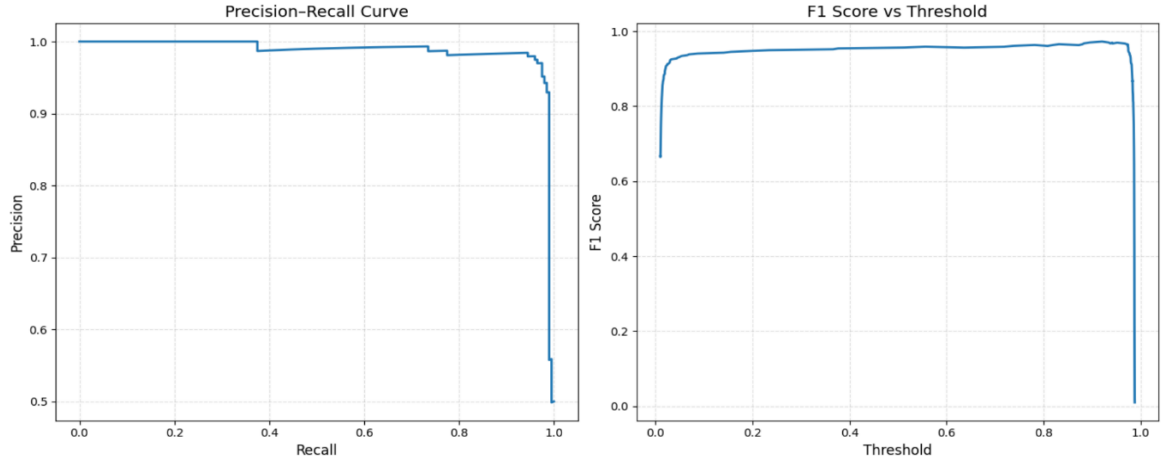
4.3.4 Precision-Recall and F1-vs-Threshold Analysis:

Further to ROC evaluation, Precision-Recall (PR) graphs and F1-Threshold graphs were used in order to obtain a better medical evaluation of the proposed quantized hybrid CNN-ViT models, particularly in the case of an imbalanced dataset, which is common in medical image data. While ROC curves can sometimes be overly optimistic in favor of negative instances dominating, PR curves consider only the positive instance, making them more relevant in cancer detection where false negatives have major clinical implications [41].

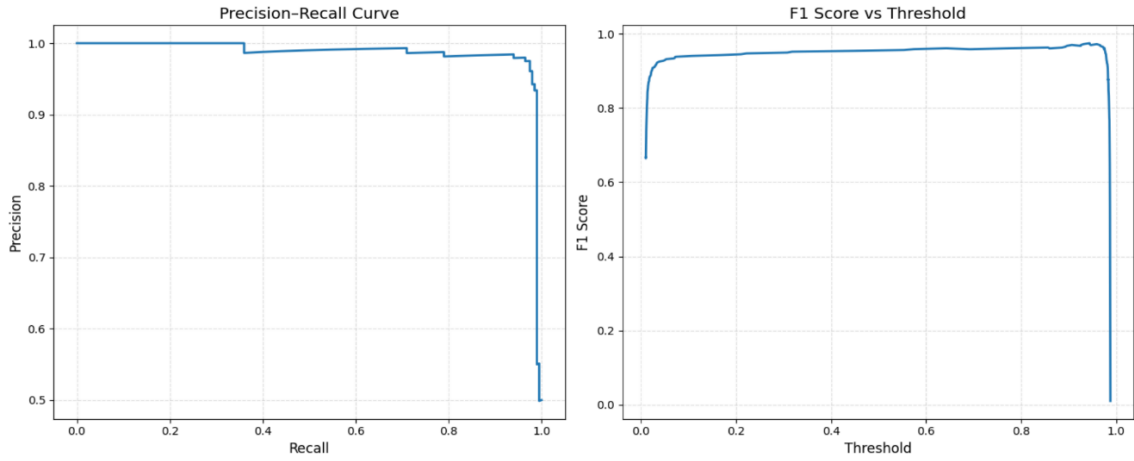
The PR curve demonstrates the trade-off between precision and recall at different thresholds. Recall ensures that the test picks up all the positive cases, while precision reduces the number of false positives picked up by the test. The balance between recall and precision is important in a diagnostic pathway because it determines the rate of false negatives and false positives.

The F1 vs Threshold graph further helps determine the most appropriate operating threshold through an analysis of how the F1 score changes as a function of shifting boundaries. The reason for this is that the clinical use of the algorithm necessitates the selection of one global threshold. The F1 score can be maximized at one particular threshold point to achieve both high sensitivity and reduced false positives.

Taken altogether, both threshold analyses provide a detailed insight into the performance of our model, supplementing results obtained via ROC-AUC.



(a)



(b)

Figure 4.4: Precision vs Recall plot and F1 score as a function of the decision threshold for the proposed CNN-ViT hybrid quantized model. In Figure (a), the Precision vs Recall curve is presented, while in Figure (b), the variation of F1 score with respect to the threshold is shown.

4.3.5 Model Size Reduction Through Post-Training Quantization:

In order to investigate whether the proposed hybrid model of CNN and ViT can be deployed on resource-limited devices, two types of post-training quantization methods were implemented, including Dynamic-Range Quantization (DQR) and Float16 Quantization. The baseline FP32 model occupies 26.78 MB, which is relatively common among hybrid models based on transformer-CNN but not suitable for tinyML scenarios.

The quantization process drastically cut down on the storage space used by the model while still achieving accurate predictions. DQR was found to be most effective since it compressed the model to just 2.38 MB, which is a 91% decrease from the original size. Similarly, float16 quantization compressed the model to 4.48 MB, resulting in an 83% decrease.

Both quantization approaches lie under the 5 MB limit typically needed for deployment in TinyML applications. This proves that the suggested model satisfies one of the key aspects of the paper, which is to develop a clinically efficient deep learning workflow that can be deployed in constrained environments.

Model Variant	Size (MB)	Reduction Relative to FP32
FP32 (Full Precision)	26.78	--
Dynamic-Range Quantization (DQR)	2.38	~91% reduction
Float16 Quantization	4.48	~83% reduction

Table 4.2: Model size comparison before and after quantization

4.3.6 Quantitative Performance Comparison of Base and Quantized Models

To prove the effectiveness of the hybrid architecture designed, an elaborate comparison of its performance was conducted against a number of other benchmark and fusion-based models with the use of an identical test dataset. Among the models for comparison were lightweight pretrained CNNs, such as MobileNetV2 and MobileNetV3Small; those models were chosen for their appropriateness in resource-limited settings. In contrast, MobileNetV3Large was not considered as an option owing to its high amount of parameters and memory usage, which is contradictory to the TinyML concept. Besides, a custom vision transformer (ViT) architecture was used to examine the role of global self-attention independently. Further, for the evaluation of the hybrid approach itself, a hybrid MobileNet-ViT model was created, which uses extracted

features from MobileNetV3Small and a custom ViT as inputs into a custom classification head. The baseline models facilitate a stringent comparison with the new hybrid model architecture. The model performance has been evaluated based on accuracy, area under the curve (AUC), precision, sensitivity/recall, F1 score, specificity, and model size. Additionally, the performance metrics have been tested with two different post-training quantization methods- dynamic range quantization (DRQ) and float16 quantization (F16)-in order to test their effectiveness under compression.

The below table depicts the exact quantitative analysis performed for this experiment. It is evident from the results obtained that CNN-ViT outperforms other models when it comes to achieving a balance between classification accuracy and resistance to quantization.

Model	Quantization	Accuracy	AUC	Precision	Recall	F1-score	Size (MB)
CNN-ViT Hybrid	Baseline	96.5%	98%	95%	98%	96%	26.78
	DRQ	97%	98%	97%	97%	97%	2.38
	F16	96.75%	98%	95%	98%	96.79%	4.48
MobileNet-ViT Fusion	Baseline	94.5%	97.89%	94.5%	94.5%	94.5%	13.31
	DRQ	95.5%	97.64%	97.39%	93.5%	95.4%	2.71
	F16	94.75%	97.99%	94.99%	94.5%	94.74%	4.91
MobileNetV3Small	Baseline	94.5%	97.65%	94%	95%	94.53%	11.31
	DRQ	94%	97%	96.88%	91%	93.81%	1.06
	F16	93.75%	97.76%	93%	94.5%	93.71%	1.82
Custom ViT	Baseline	93%	96.89%	92%	94%	93%	17.11
	DRQ	94.25%	96%	94.72%	94%	94.24%	1.53
	F16	93%	96.89%	92%	94%	93%	2.87
MobileNetV2	Baseline	93.25%	97.72%	92.61%	94%	93%	26.21
	DRQ	92.25%	98%	91.95%	93.5%	92.35%	2.39
	F16	93%	97.66%	92%	94%	93%	4.26

Table 4.3: Comparison of baseline models' performance and their quantized counterparts (DRQ and F16), with the following metrics being reported: Accuracy, AUC, Precision, Recall, F1 Score, Specificity, and Size. It is evident that the novel CNN-ViT model has the most optimal trade-off in terms of its accuracy-efficiency profile.

Discussion on the comparison table: Discussion of the comparison table: As can be seen in the comparison table, the CNN-ViT hybrid model surpasses its competitors based on most discrimination criteria while maintaining its robustness with a baseline accuracy of 96.50% and an AUC value of 98%. With DRQ, the model's accuracy goes up to 97.00% while its size decreases significantly from 26.78 MB to 2.38 MB. In addition, the float16 (F16) version of the model provides almost identical results in terms of performance (accuracy = 96.75%, size = 4.48 MB). While quantization usually aims at compressing the model and lowering inference costs, some recent works have demonstrated that quantization acts as an implicit regularization technique that maintains or slightly enhances accuracy. It allows quantized neural networks to converge to flatter minima of the loss surface, improving their generalization capabilities compared to their full-precision equivalents. Efficient deployment literature also suggests similar findings. Moreover, advanced quantization techniques have proven to be lossless in applications related to medical imaging, implying that reducing precision not only does not negatively impact prediction accuracy but might also improve it [42][43].

In conclusion, the CNN-ViT fusion model shows outstanding discriminatory capability (accuracy & highest AUC), retains high class balance (precision and recall), as well as robustness towards aggressive model compression; these traits combined make the CNN-ViT fusion the most ideal model choice for deployment post quantization.

4.4 Explainable AI (XAI) Integration

To improve the interpretability and clinical validity of the suggested CNN-ViT hybrid architecture, an explainable AI (XAI) module was incorporated using the LIME technique (Local Interpretable Model-agnostic Explanations). The role of the XAI module is to generate visual interpretations of the classifier output by indicating the portions of the MRI images used in making the decision [44].

This kind of integration is particularly important for healthcare applications as this would make sure that the AI model uses relevant information and not any spurious artifacts to classify the image or report. Additionally, XAI techniques can help to provide information about the model's working in both correct and incorrect classifications, thus increasing user confidence and clinical workflows in the AI tool.

The sample interpretations generated by XAI for five testing instances are illustrated in Figure 9. The selected zones demonstrate the zones where the CNN network and ViT network have detected texture patterns and morphology patterns, respectively.

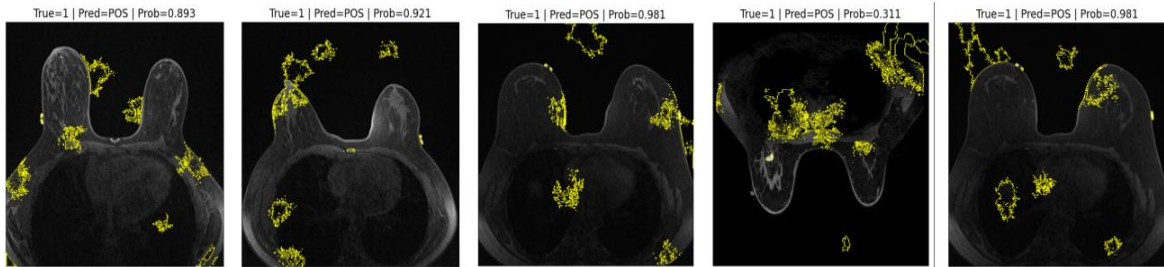


Figure 4.5: XAI interpretation on five representative MRI slices based on LIME. The marked regions show the parts that contribute the most to the predictions made by the CNN-ViT hybrid model

Chapter 5. Conclusion

5.1 Discussion

This paper proposes an innovative CNN-ViT model for detecting breast cancer using pre-contrast T1-weighted DCE-MRI images. It is found that the CNN-ViT model outperforms existing approaches, especially in identifying minute malignant structures. This is due to the fact that the CNN model extracts local structural features, while the ViT model captures long-range correlations between different tissue components.

The proposed model performed better than CNN and ViT standalone baseline models and the MobileNet-ViT fusion model by giving very high accuracies, sensitivities, and AUCs without suffering from any overfitting. Considering that overfitting did not occur although the model was deep suggests that the convolution down-sampling mechanism and token representation with multihead self-attention complement each other well.

One significant achievement of this study is the ability to show robustness in post-training quantization. Quantized models using both dynamic range and Float16 quantization showed near-identical levels of prediction accuracy with a reduced model size to 2.38MB and 4.48MB, respectively. This drastic reduction in model size without any decline in diagnostics proves its feasibility in TinyML where memory, computational power, and energy consumption are constrained.

Interpretability evaluation using LIME demonstrated consistent attention towards areas adjacent to tumors and boundaries of clinically significant tissue structures, while ensuring interpretability in quantized form. Classification errors have been mostly observed in difficult and low-contrast cases, which pose problems even to radiologists. On the whole, the introduced CNN-ViT architecture demonstrates excellent trade-offs between precision, interpretability, and efficiency, making it an excellent fit for resource-constrained environments.

5.2 Limitations and Future Work

However, several limitations must be noted despite the successful results presented by the suggested hybrid CNN-ViT architecture. Firstly, the LIME-based explainability technique sometimes highlights non-relevant areas outside the tumor region. While most explanations provided by the algorithm tend to concentrate on relevant parts, such issues demonstrate that post-hoc methods have their limitations in terms of interpretability. Therefore, such an approach can only be seen as an addition rather than an independent solution for clinical applications. Secondly, even though post-training quantization significantly downsizes the model by achieving less than 5 MB after optimization - thus allowing deployment in the context of TinyML, additional optimization should be carried out for use in ultra-low memory systems. Thus, building an effective model with less than a megabyte in terms of memory becomes a necessity.

As far as future work is concerned, there are multiple avenues for further exploration that arise from this study. For example, extending the framework proposed here to multi-modal imaging cases, such as fusing MRI images with other imaging modalities like ultrasound or mammography, could lead to better results because of the additional information being leveraged. Another avenue of future work would be incorporating advanced techniques of explainability into the model, for instance, using integrated gradients, attention visualization, or transformers.

Lastly, investigating other approaches to model compression other than post-training quantization, such as structured pruning and knowledge distillation, can help achieve memory savings without sacrificing accuracy. This would enhance the viability of implementing hybrid CNN and Transformer models in actual medical settings.

Overall, resolving the above challenges and exploring the proposed future research directions would help in enhancing the robustness of hybrid CNN-ViT architectures for clinical applications.

5.3 Clinical Implications

The presented CNN-ViT hybrid approach has many useful clinical applications regarding breast cancer detection on pre-contrast MRIs. High accuracy and consistency among the various versions of quantization show that the method is well applicable in clinical environments. It shows flexibility since it does not require any architecture adjustment to be used with MRI images. From a functional point of view, the presented model can work as an intelligent second reader helping to find malignancies more precisely and reduce interobserver variability typical for the diagnosis of breast cancers. Besides, thanks to LIME, the hybrid model is able to generate saliency maps that show which parts of the image influence its predictions. Such explainable AI techniques increase the interpretability of the approach and help the clinicians to check how the model makes its decisions, which is an essential advantage of the methodology. Thus, the CNN-ViT hybrid framework can become a helpful solution in breast cancer detection due to its clinical value, explainability, and applicability.

5.4 Conclusion

A novel CNN-ViT hybrid architecture was proposed in this study for breast cancer detection based on pre-contrast T1-weighted DCE-MRI slices. This model exploits the power of convolutional layers for capturing local features of images and Vision Transformers for their global understanding to circumvent the limitation of using only CNNs or transformer-based models in analyzing high-resolution medical images. Through local texture cues and long-range structures of MRI images, this CNN-ViT hybrid model can provide an efficient discrimination of image features relevant to early-stage tumors.

It offers a small number of experiments on the MRI slice-level dataset where the performance of the suggested model is evaluated against multiple evaluation criteria, including accuracy, AUC, precision, recall, F1 score, and specificity. Most notably, the model's performance remained consistent during quantization following training, with Float16 and dynamic-range quantization offering similar diagnostic capabilities while decreasing the model size by a factor of four, from 26.78 MB to 2.38 MB, respectively. This information may be used to confirm the appropriateness of using the hybrid model for TinyML in clinical environments.

Also, through explainability analysis using the LIME, it was established that the model considers all areas of the body that are of clinical significance. This kind of understanding is vital for determining the transparency of the medical AI systems as well as their level of trustworthiness and demonstrates that quantization did not affect the key decision-making principles of the model.

In general, the proposed CNN-ViT model architecture is an effective, interpretable, and lightweight network which can help in detecting breast cancer through MRI images. Considering its efficiency without any compromise in its diagnostic capacity even under significant compression, it makes an excellent training process for future medical diagnostic devices. Possible applications in the future could include the patient-level prediction, MRI image sequences fusion, and prediction optimization from the point of view of hardware.

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